# GH3036 & GH3038 EVK PC Tool User Manual

**Revision: 0.2**

**Release Date: 2025-03-06**

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# Preface

## Purpose

This article describes the external specifications, basic usage and principles of the GH3036 & GH3038 evaluation kit PC tool (hereafter referred to EVK PC Tool), which will help users quickly familiarize themselves with the platform environment and speed up the evaluation and development process.

## Audience

This article is intended for the following readers:

- GH3036 & GH3038 users
- Software engineers
- Test Engineers

## Release Notes

This document is the first release, and the corresponding software version is EVK PC Tool V02.02.03.

## Revision History

| Revision | Date | Description |
|---|---|---|
| 0.1 | 2024-12-11 | Preliminary release |
| 0.2 | 2025-03-06 | Updated the document format and added Section 8.4.7 |

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# 1 Overview

The GH3036 & GH3038 EVK PC Tool is designed for GH3036 & GH3038 sensor performance evaluation and application development.

This document focuses on software functions and operation instructions for this kit, including:

(1) Install and Launch software;

(2) Connect to EVK board;

(3) Start sample with selected config file;

(4) Evaluate data in graphic and file;

(5) Specifications of EVK software tool.

While excluding:

(1) Configuring parameters, see <*GH3036 & GH3038 Customer Project Parameter Adjustment Guide*>.

(2) EVK Board hardware, see *<GH3036 & GH3038 EVK Quick_Start_Guide>*.

Before reading this document, please make sure you have read <*GH3036 & GH3038 EVK Quick_Start_Guide*>.

Figure 1-1 GH3036/GH3038 EVK Board Kit

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# 2 Software Installation

## 2.1 Overview

Launch the EVK PC Tool installer (GH3036&GH3038EVK_setup.exe and GH3036&GH3038_QtDLL_setup.exe), and take the following steps to install and launch the software.

1. Launch *GH3036&GH3038EVK_setup.exe,* click Browse and select the installation location, then click Next;

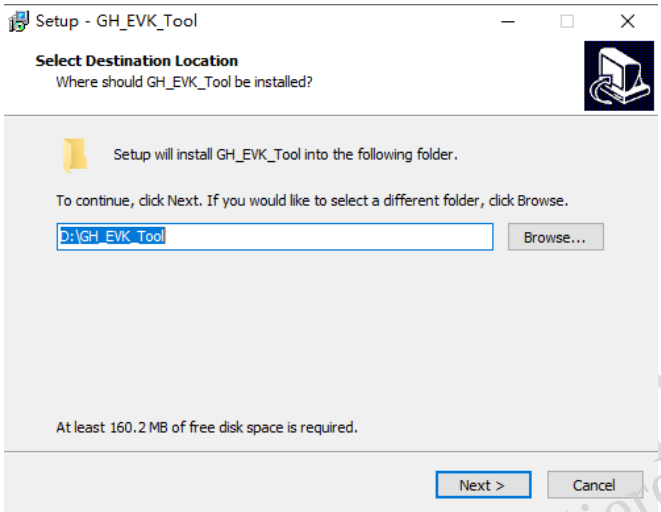


Figure 2-1 Select the installation location

2. Keep the default Start Menu Folder unchanged, then click Next;

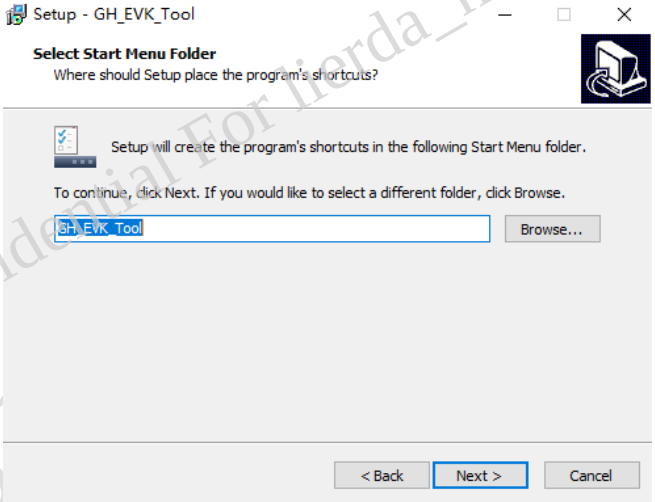


Figure 2-2 Select Start Menu Folder

3. Determine whether to create a desktop shortcut and install the drivers automatically as required, then click Next.

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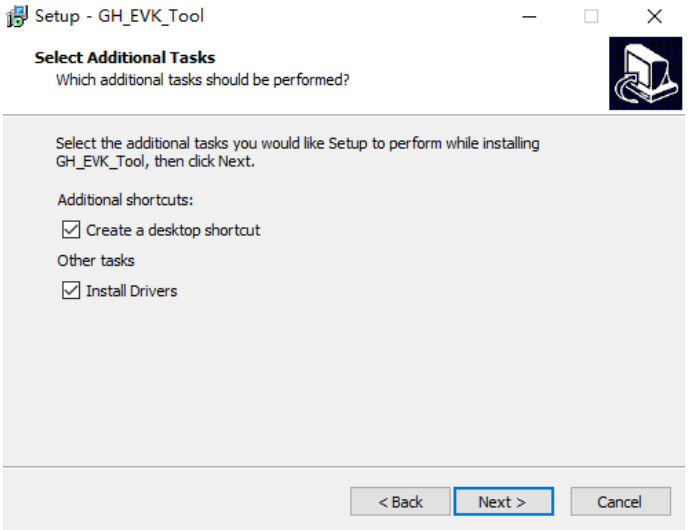


Figure 2-3 Select Additional Tasks

4. Click Install to start the installation process.

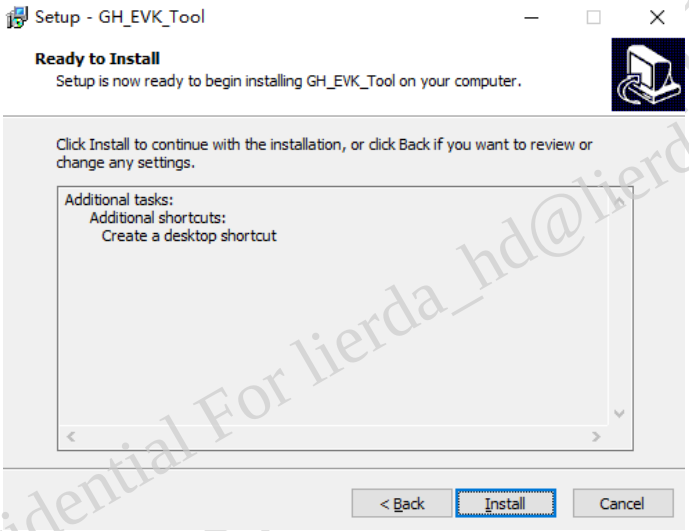


Figure 2-4 Software Installation

5. When the installation is complete, click Finish to complete the installation process and open the EVK PC Tool.

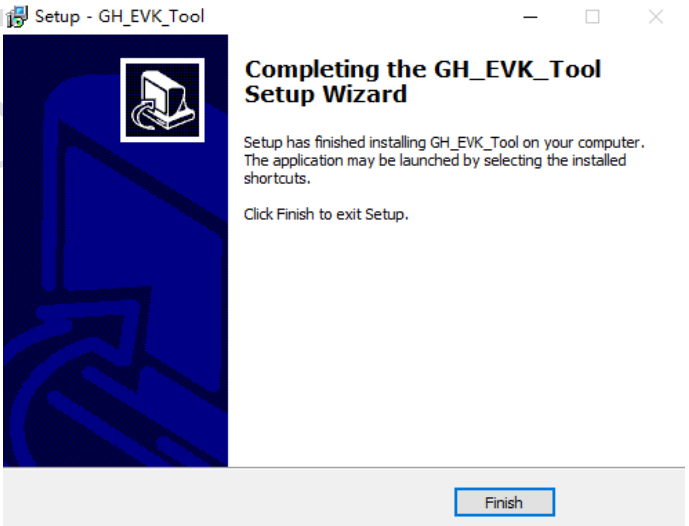


Figure 2-5 Installation Complete

6. If your computer does not have the QT environment installed, you will need to additionally install the QT dll libraries by running the GH3036&GH3038_QtDLL_setup.exe.

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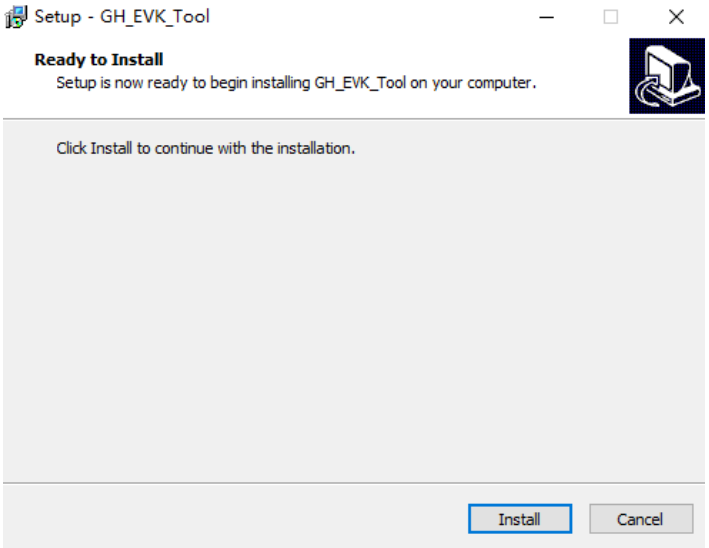


7. Select the chip you need, then click Ok to enter the main interface of EVK PC Tool.

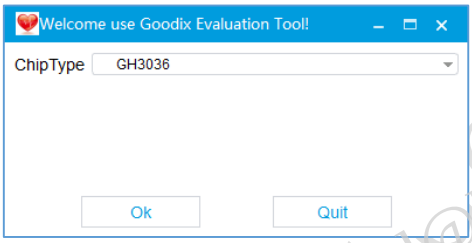


Figure 2-6 Select the chip

## 2.2 Driver Installation Selection

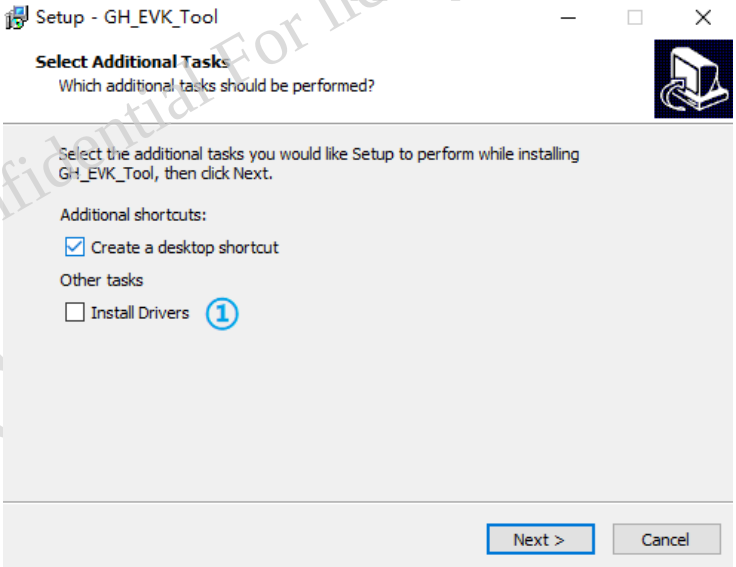


Figure 2-7 Install Drivers

Table 2-1 Additional task manual

| UI ID | Type | Description | Note |
|---|---|---|---|
| ❶ | CheckBox | Install Serial, BLE dongle Serial and C++ runtime library | Checked by default only during the first installation.<br>If there is a driver installation error or the system is Windows 7, please refer to Table 8-3 Known issues. |

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## 2.3 Login Page

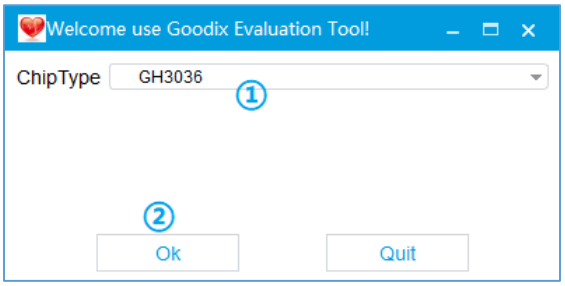


Figure 2-2 Login page

Table 2-2 Login Page manual

| UI ID | Type | Description | Note |
|---|---|---|---|
| ❶ | ComboBox | Select the required chip model for verification. | - |
| ❷ | Pushbutton | Confirm selected chip model and enter application | After clicking OK, no modifications can be made anywhere unless the application is restarted |

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# 3 Main Interface

## 3.1 Overview

The main interface consists of some bars, which are title bar, switch bar, and status bars, as shown below.

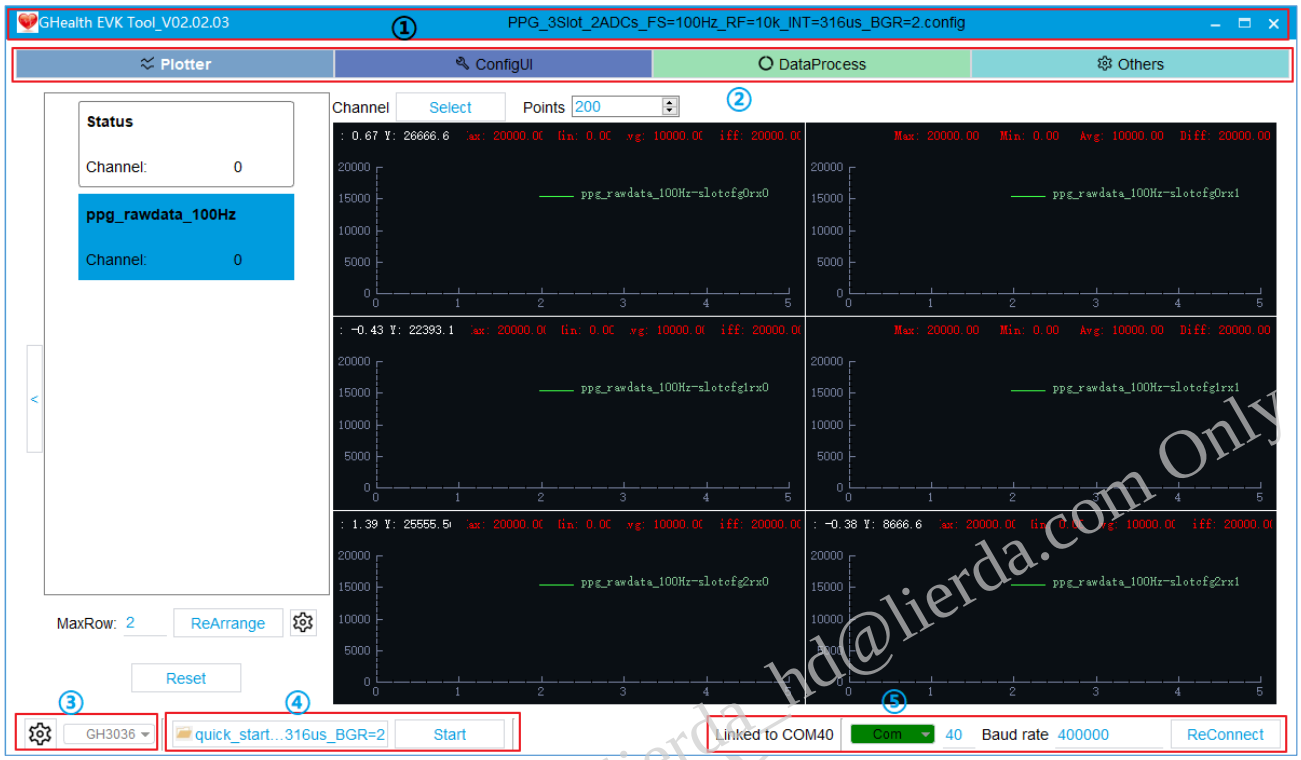


Figure 3-1 Main Interface

1. The Title bar shows the name of the config file which is loaded now and will vary if the config file changes.
2. The switch bar displays the four modules of the EVK PC Tool. Users can show multiple modules at the same time by double-clicking on the desired module.

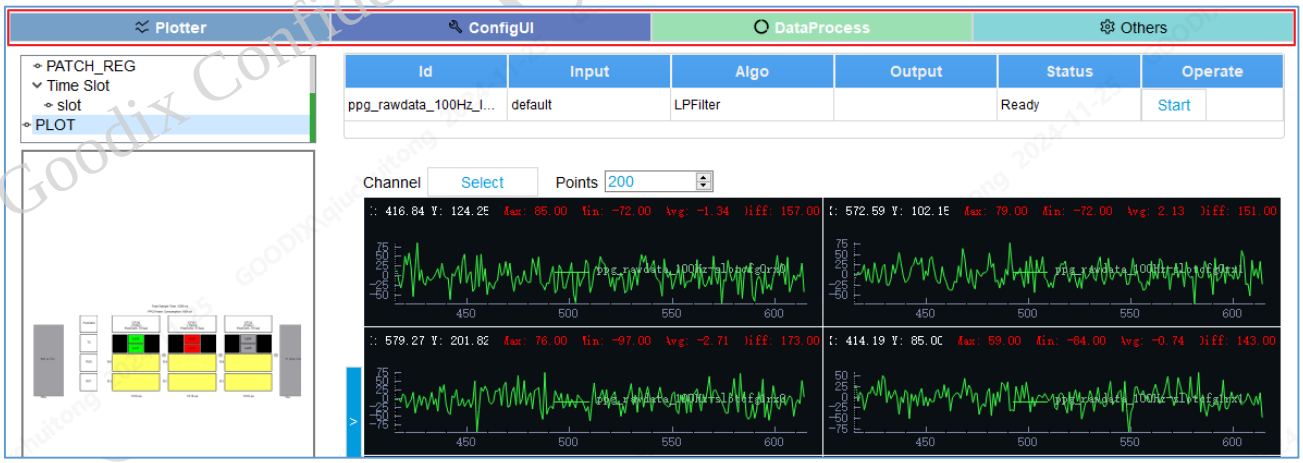


Figure 3-2 Switch bar

Table 3-1 Function page manual

| Module | Description |
|---|---|
| Plotter | Show data uploaded by MCU, and save file. |
| ConfigUI | Create, modify and send config file. |
| DataProcess | Check filter and algorithm status. |
| Others | Interact with MCU. Including firmware update, register reading/writing. |

3. The version bar displays the selected chip on the login page. By clicking on the icon ⚙, users can check the versions of the components.

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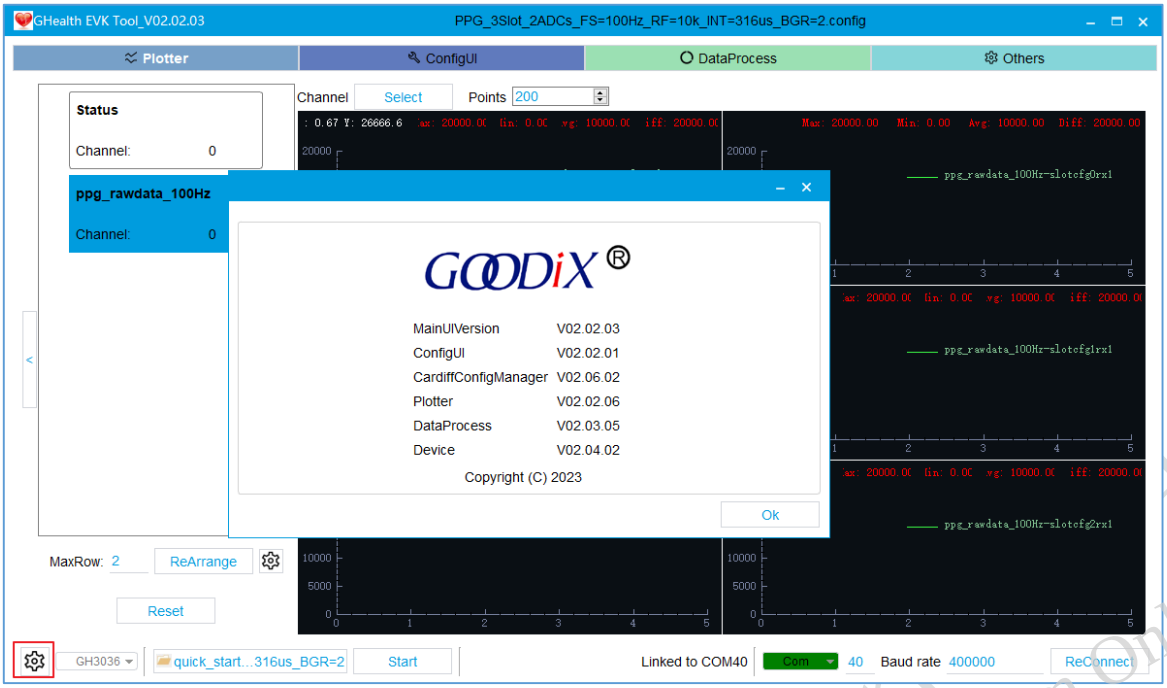


Figure 3-3 Version bar

4. The config bar shows the name of config file you have selected, and users can click config switch to select the other config files.

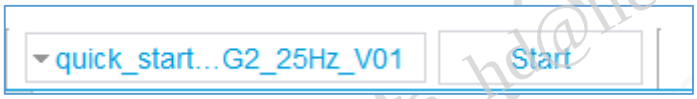


Figure 3-4 Config bar

5. The connection bar displays the connection mode and status. When the EVK PC Tool is connected to the EVK Board using a BLE dongle or USB cable, please wait for the device status bar to show the serial COM ID or BLE address of the EVK board.

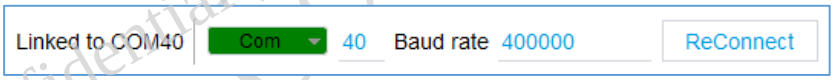


Figure 3-5 Linked to EVK board by serial port

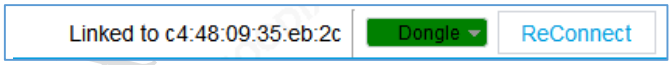


Figure 3-6 Linked to EVK board by BLE dongle

## 3.2 Config Operation

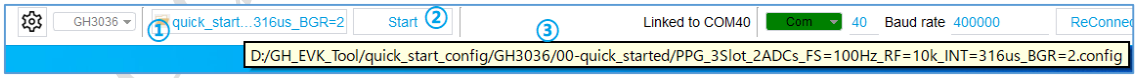


Figure 3-7 Config bar

Table 3-2 Config bar manual

| UI ID | Type | Description | Note |
|---|---|---|---|
| ❶ | Pushbutton | Show selected config file name; Select the other config file by clicking it to show 3.3 Config Selection page | If the length of the config file name exceeds the control width, a part of it will be hidden. By moving the mouse over the ID❶, user can see the complete config file name.<br>The config selection page hides when it loses mouse focus (such as when mouse clicks other window or other window shows up). |
| ❷ | Pushbutton | Start/Stop sampling | The config content is checked when starting the sampling process. If any issues are detected and the user refuses to load and apply the configuration compulsively, the sampling will be automatically stopped. |

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## 3.3 Config Selection

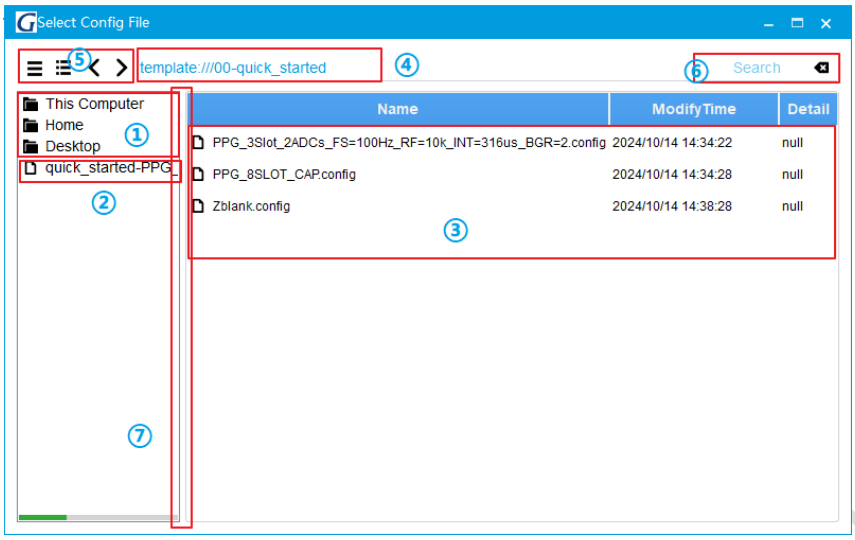


Figure 3-8 Config selection page

Table 3-3 Config selection manual

| UI ID | Type | Description | Note |
|---|---|---|---|
| ❶ | List item | Quick path | - |
| ❷ | List item | The config files opened recently | It can be opened quickly with a single click |
| ❸ | List | Config file or dictionary | - |
| ❹ | Lineedit | Path | Support editing |
| ❺ | Pushbutton list | Operations about path and display | The forward button only supports navigating back to the previous path once. |
| ❻ | Lineedit | Search files or folders quickly | Do not distinguish between files and folders |
| ❼ | Spliter | Adjust the width of the shortcut menu using the mouse | - |

## 3.4 Device Connection

The EVK PC Tool can connect to EVK board by Serial Port and Bluetooth.

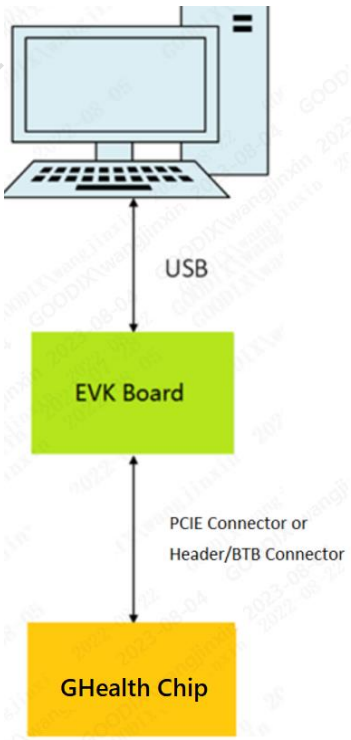


Figure 3-9 Serial connection structure

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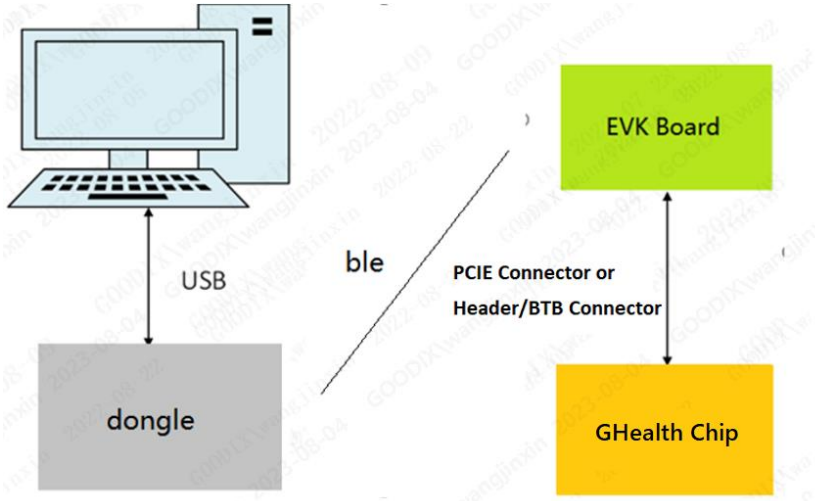


Figure 3-10 Bluetooth connection structure

### 3.4.1 Bluetooth Connection

Take the following steps to connect to Bluetooth:

1. Plug BLE dongle into the PC and wait for the connection page to pop up.

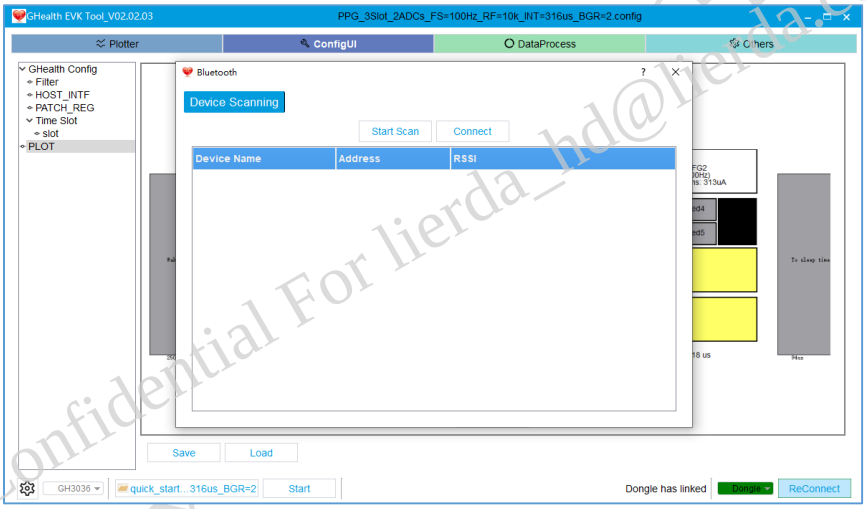


Figure 3-11 Bluetooth Connection Interface

2. Click **Start Scan** button to collect information of surrounding devices, the button becomes **Stop Scan** after scan process begins.

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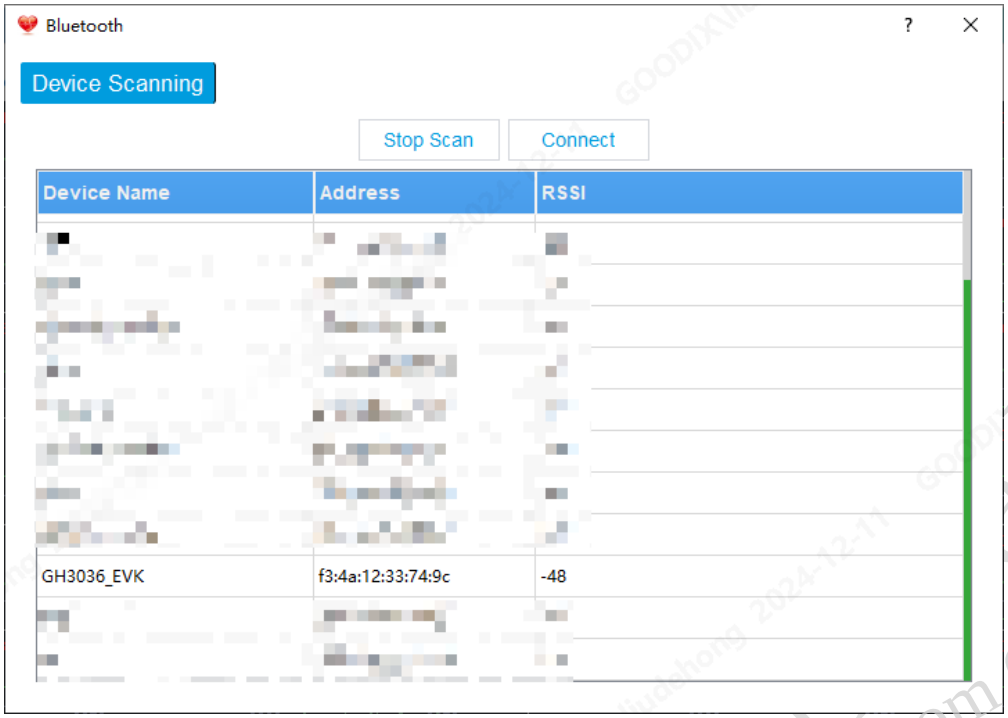


Figure 3-12 Start scan device

3. Select the Bluetooth address corresponding to the device which is named **GH3036_EVK** and click **Connect**.

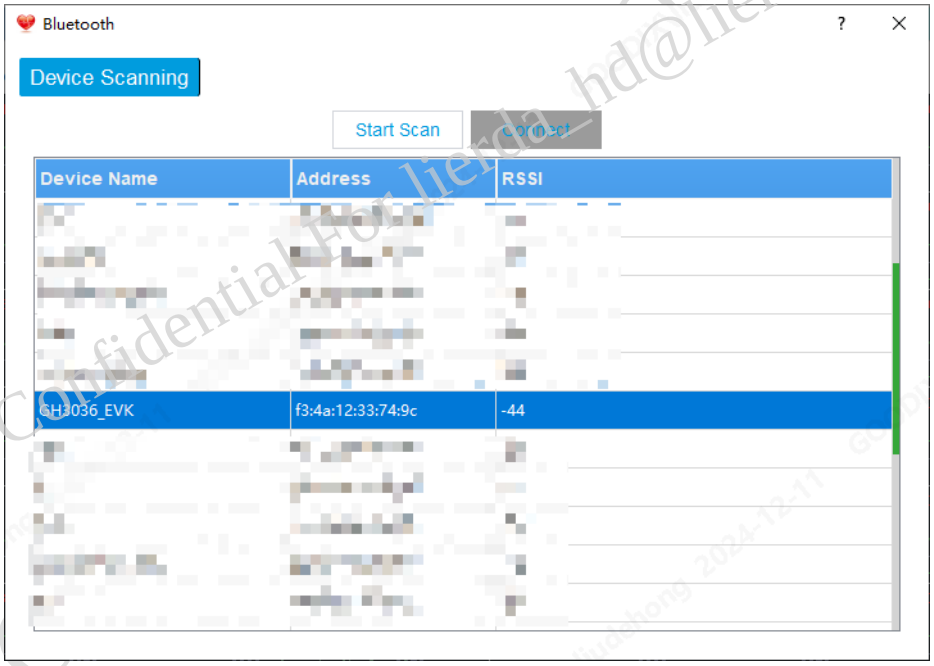


Figure 3-13 Connect selected device

4. Wait until the connection page exits automatically, and the Bluetooth MAC address is shown in the lower right corner of the main interface of the EVK tool if the Bluetooth is connected successfully.

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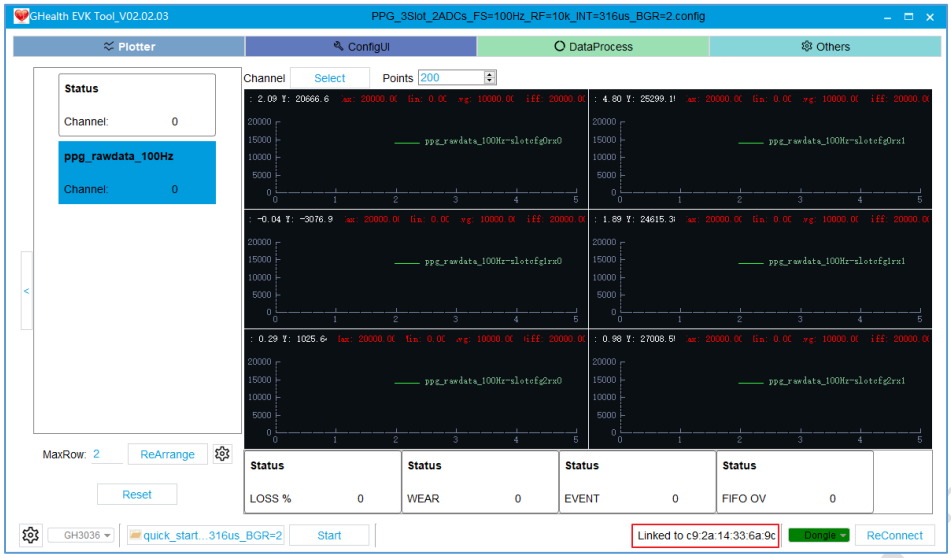


Figure 3-14 Status Display of Bluetooth Communication

## 3.4.2 Serial Connection

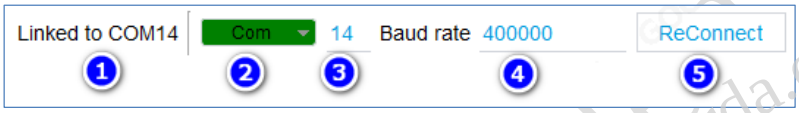


Figure 3-15 Connection bar

Table 3-5 Connection bar manual

| UI ID | Type | Description | Note |
|---|---|---|---|
| ❶ | Label | Show connection status | 1.**Linked to xxx** means the connection was successful<br>2.Warning shows up if you attempt to operate the device when it is not ready |
| ❷ | ComboBox | Connection type, use serial (COM) by default | Only take effect after clicking Reconnect;<br>When dongle is selected, id3 and id4 will become hidden, please refer to section 3.4.1 Bluetooth Connection |
| ❸ | Lineedit | Com ID | Hidden when dongle is selected as connection type |
| ❹ | Lineedit | Baud rate of serial port | Hidden when dongle is selected as connection type |
| ❺ | Pushbutton | Reconnect to EVK board by selected connection method | - |

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# 4 Plotter

## 4.1 Overview

Plotter provides functions for data visualization and saving.

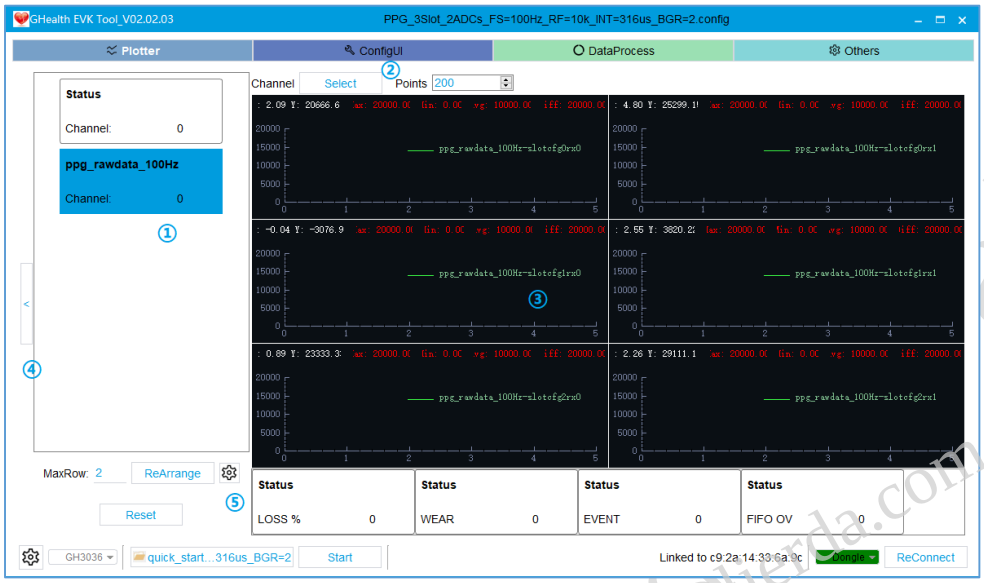


Figure 4-1 Plotter

1. Present all the received channel data and organize it into a file frame. Execute operations with this file by using the right click menu.

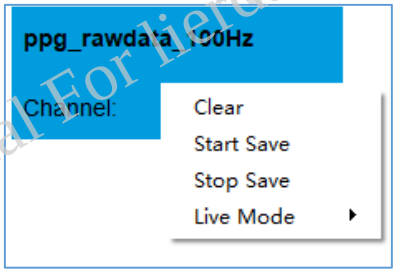


Figure 4-2 File Right clicking menu

2. The channel selection dialog presents all channels of the selected file frame for the users to check and display as wave plots. Users can also enhance the current wave plot by enabling the controller to perform other operations on the selected channel.

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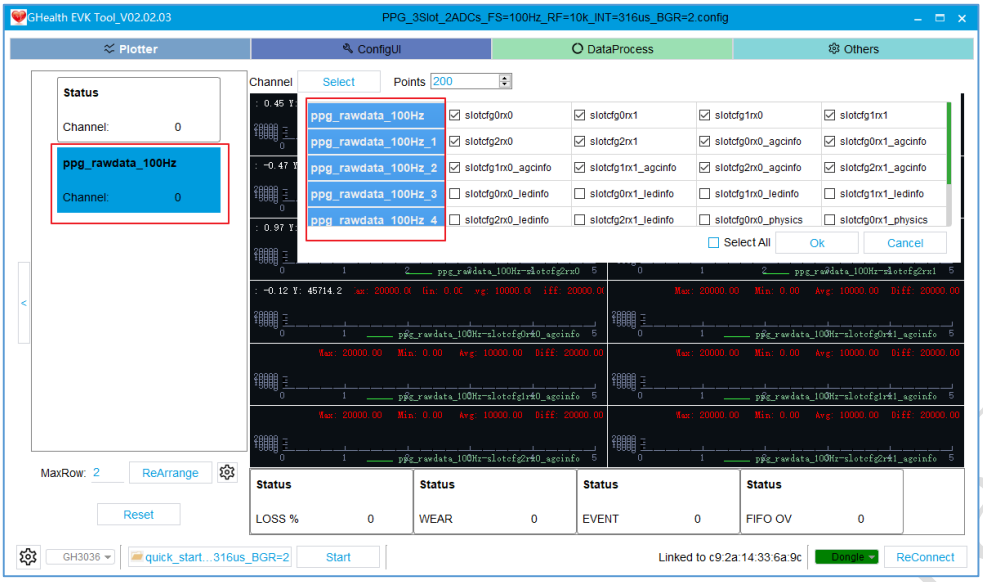


Figure 4-3 The selected file and the corresponding channel

3. Wave plot presents data by channels. Execute operations on this channel through a right-click menu or by double-clicking.

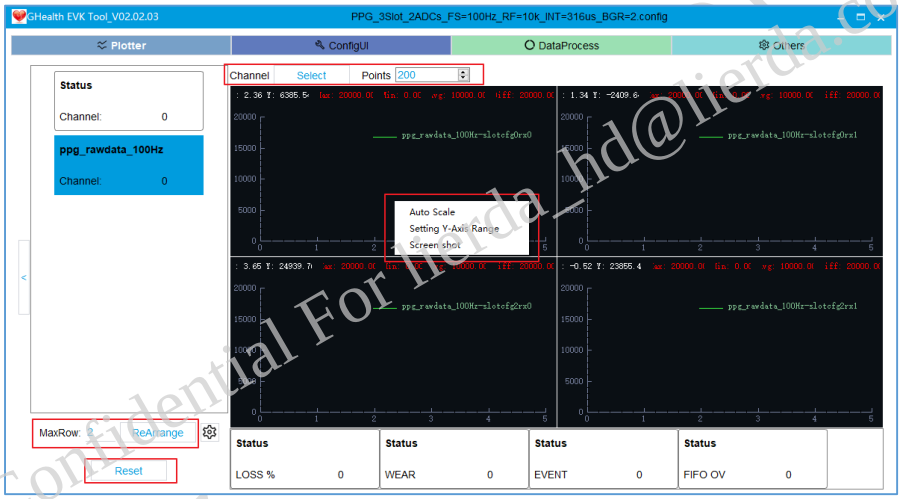


Figure 4-4 Wave plot controllers with right click menu

4. Click fold up button to let Frame and config zone fold up or unfold.

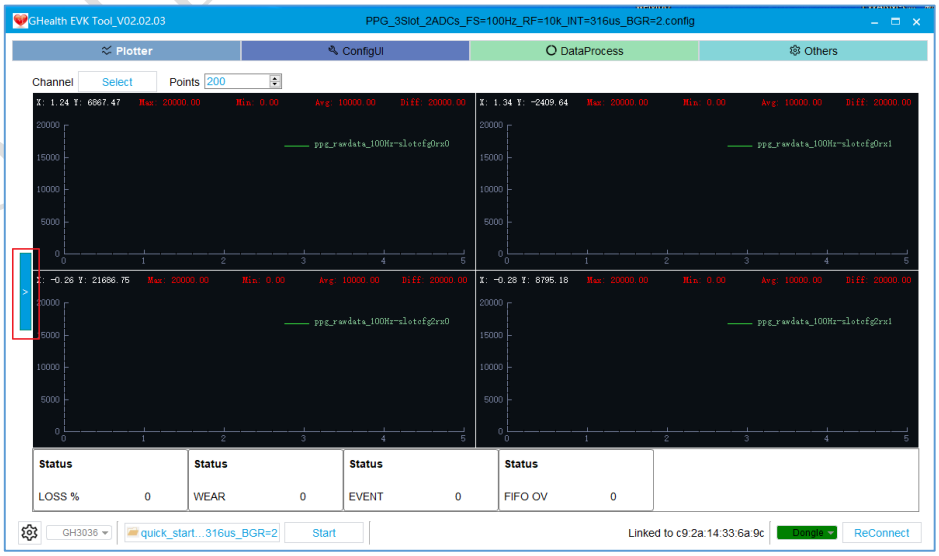


Figure 4-5 Fold button &fold mode in plotter

5. To modify the file saving mode, users can click on the file config icon button, and then the file saving mode

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dialog will open up, users can refer to section 4.4 .

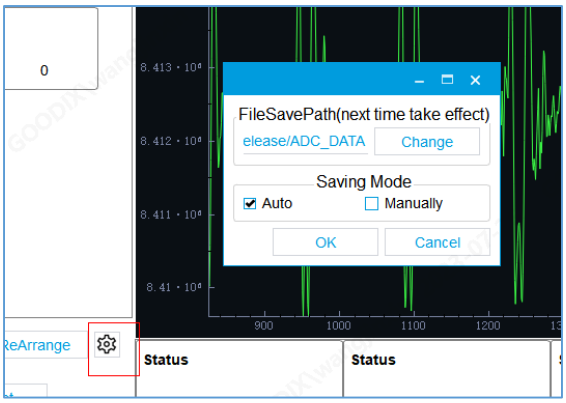


Figure 4-6 File config button& file save mode dialog

## 4.2 Waveform Plot

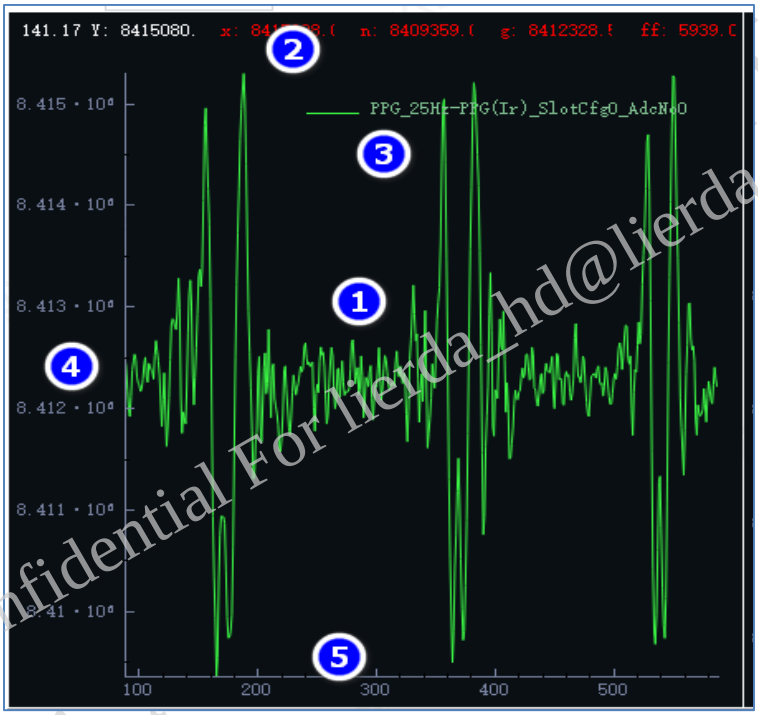


Figure 4-7 Waveform Plot

Table 4-1 Waveform plot manual

| UI ID | Type | Description | Note |
|---|---|---|---|
| ❶ | Waveform | The waveform plot drawn from the collected data | - |
| ❷ | Labels | Coordinate and statistical information | Statistical information only includes points in plot |
| ❸ | Label | File& channel information | Format is Filename_channel name |
| ❹ | Axis | Y-Axis shows value of data | - |
| ❺ | Axis | X-Axis shows point number | - |

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### 4.2.1 Operation

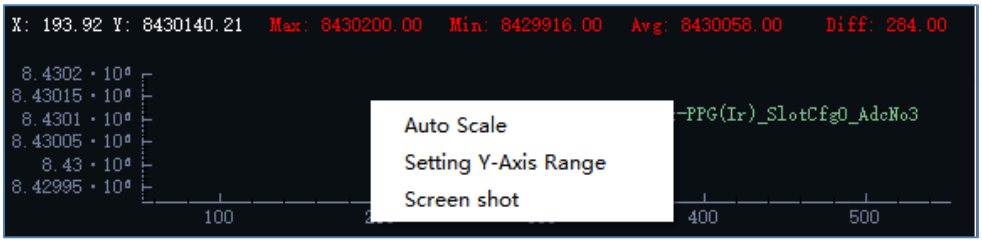


Figure 4-8 Plot with right click menu

Table 4-2 waveform plot operation

| Operation | Item | Description |
|---|---|---|
| Right Click menu | Auto Scale | - |
| Right Click menu | Setting Y-Axis Range | - |
| Right Click menu | Screen shot | Save the waveform capture of the current channel in the following folder: ScreenShots |
| Double click | View detailed waveform in a separate window | The waveform can be freely enlarged and reduced in a separate interface. Close this window when modifying the selected channel. |

Figure 4-9 Detailed waveform window

### 4.2.2 Controller

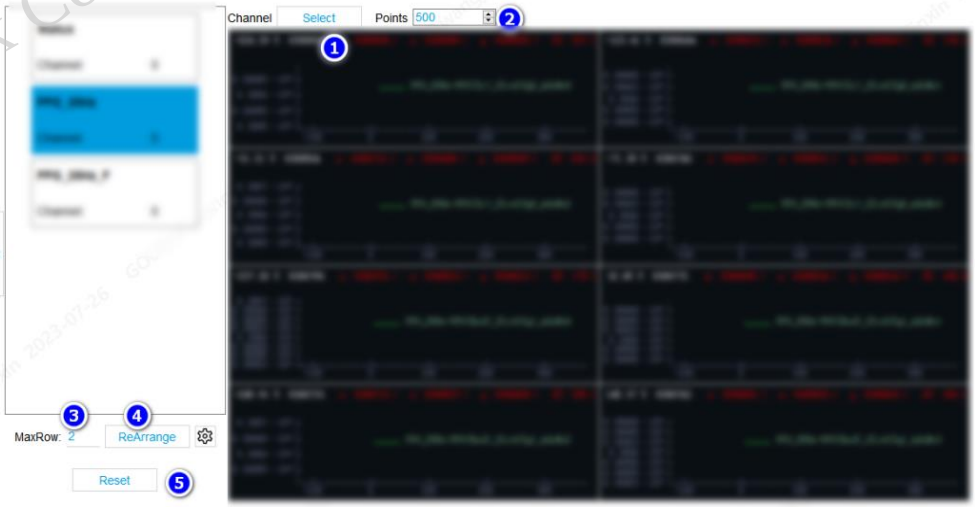


Figure 4-10 Plot controllers annotation

Table 4-3 Plot display controllers manual

| UI ID | Type | Description | Note |
|---|---|---|---|
| ❶ | Pushbutton | Show channel selection dialog | Only present the channels of selected file frame. Channel selection dialog disappears when it loses mouse focus (such as when you clicks other window or other window shows up). |

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| | | | |
|---|---|---|---|
| ❷ | Spin box | Control the horizontal coordinate range in all waveform plot | Immediately take effect |
| ❸ | lineedit | Show & set column number of plot zone | - |
| ❹ | Pushbutton | Confirm the setting of column number of plot zone | - |
| ❺ | Pushbutton | Initial plot zone | Reset any user and system record settings (such as channels set to be displayed) |

#### 4.2.2.1 Channel Selection

Figure 4-11 Channel selection dialog

Table 4-4 Channel selection dialog

| UI ID | Type | Description | Note |
|---|---|---|---|
| ❶ | Table header | Show file name | _x is not included in file name |
| ❷ | Checkbox | Select the channel that needs to be shown | - |
| ❸ | Checkbox | Select or cancel all channels | - |
| ❹ | Pushbutton | Confirm selected channels and make modifications take effect | This dialog will not disappear until you click OK or Cancel |
| ❺ | Pushbutton | Give up all modifications and return to the last selected status | This dialog will not disappear until you click OK or Cancel |

## 4.3 File Frame

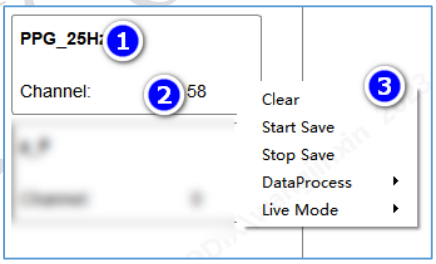


Figure 4-12 Right click menu

Table 4-5 Right click menu manual

| UI ID | Type | Description | Note |
|---|---|---|---|
| ❶ | Label | File name | Keep it the same as that of the stored file |
| ❷ | Label | The total number of channels | - |
| ❸ | Right click menu | Operations | - |

Table 4-6 Right click menu manual

| Menu | Description | Note |
|---|---|---|
| Clear | Remove this file from record | Reshow when receiving data belonging to this file |
| Start Save | Start Saving manually | Refer to section 4.4 File Saving Dialog |
| Stop Save | Stop Saving manually | Refer to section 4.4 File Saving Dialog |
| Live mode | Select a channel for Live mode, which only presents the newest data and updates once per second. | Refer to Figure 4-13 Live mode bar |
| DataProcess | Control algorithm on and off | Only configurations with algorithms will display this option. Refer to section 6 DataProcess |

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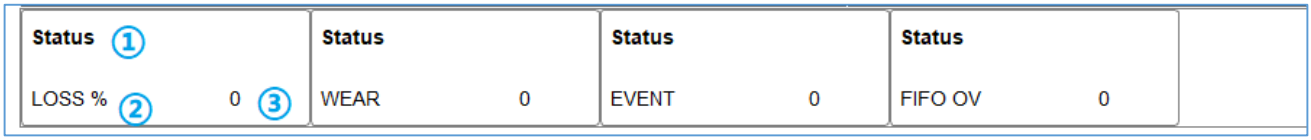


Figure 4-13 Live mode bar

Table 4-7 Live mode manual

| UI ID | Name | Detail |
|---|---|---|
| ❶ | File name | - |
| ❷ | Channel name | - |
| ❸ | Value | - |

## 4.4 File Saving Dialog

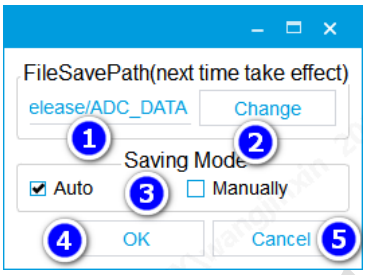


Figure 4-14 File saving dialog annotation

Table 4-8 File saving dialog manual

| UI ID | Type | Description | Note |
|---|---|---|---|
| 1 | lineedit | File saving path | This path is permanently effective |
| 2 | Pushbutton | Use Windows File Manager to change the file saving path | - |
| 3 | Check box group | Automatically: Save all data received.<br>Manually: Select when starting and stopping file saving manually | The Save button appears when 'Manually' is selected. The saved folder is named with the sampling start time |
| 4 | Pushbutton | Confirm & keep saving path and mode | Keep for a long period |
| 5 | Pushbutton | Give up all modifications and close dialog | - |

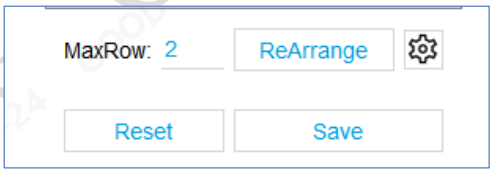


Figure 4-15 Manually save button

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# 5 ConfigUI

## 5.1 Overview

ConfigUI provides functionality for editing configuration files, and all configuration related operations are performed in this module.

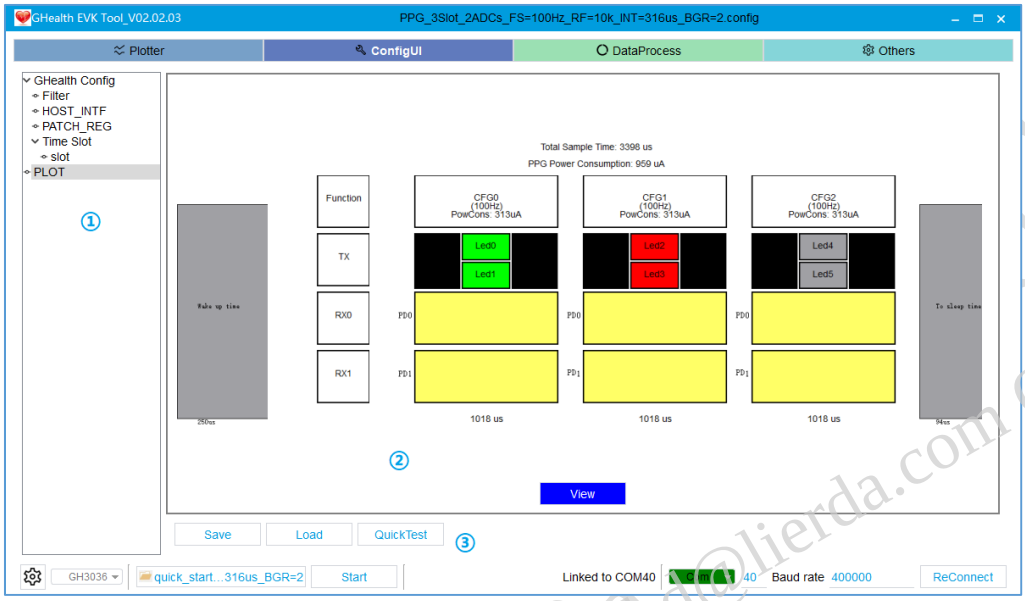


Figure 5-1 ConfigUI

1. Switch between config pages through outline and enter corresponding topic pages, then complete changes to specific aspects of the configuration; please refer to section 5.2 Outline for details. Specifically, user can use Plot page to quickly adjust the slot series or modify PPG slot config by PPG wizard; user can refer to section 5.5 PLOT for details.

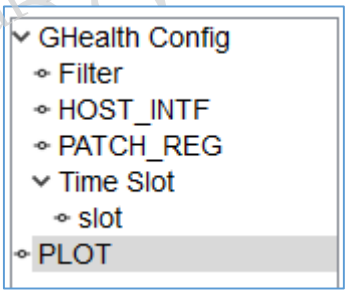


Figure 5-2 Outline

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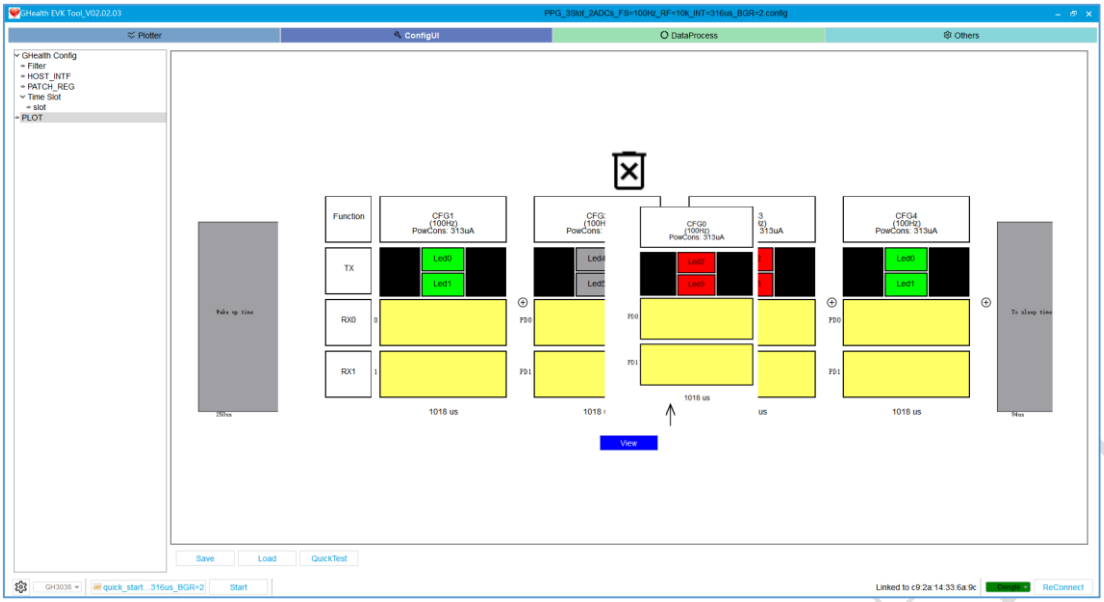


Figure 5-3 Plot page (in operation)

2. You can modify the configuration items by using the controls, which are the basic elements of a four-level hierarchy.

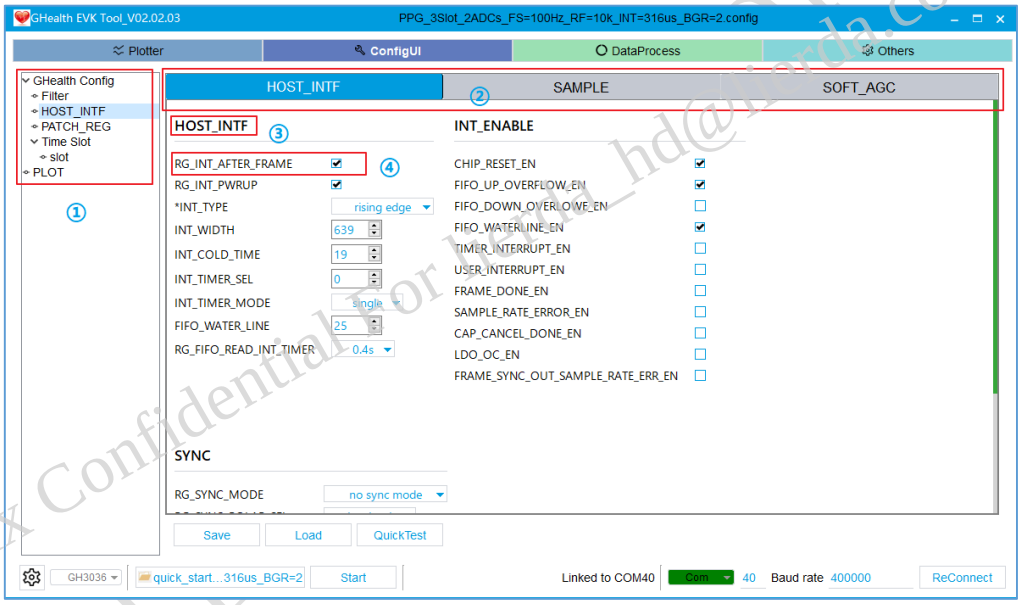


Figure 5-4 ConfigUI normal page

Table 5-1 Config page manual

| UI ID | Name | Detail |
|---|---|---|
| ❶ | Outline | Ordered topic tree |
| ❷ | Class tab | Class config to server page |
| ❸ | Group label | Class config to server group |
| ❹ | Config item | Refer to section 5.4 Controls |

3. Load or save config file by windows file manager, or launch current config quickly by clicking QuickTest; user can refer to section 5.3 Config Operations.

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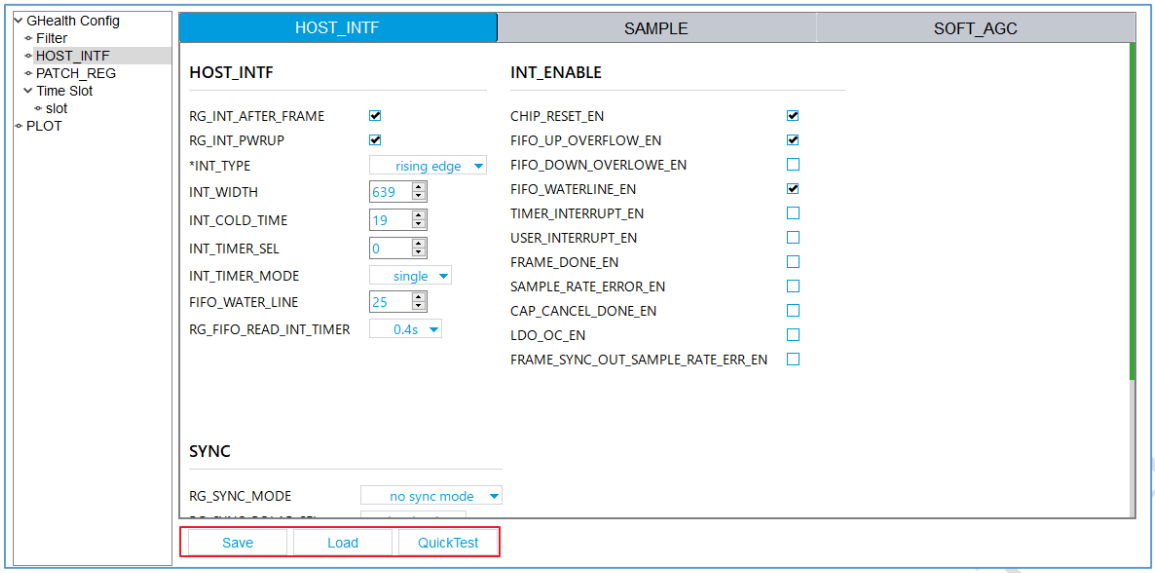


Figure 5-5 Config file operations

## 5.2 Outline

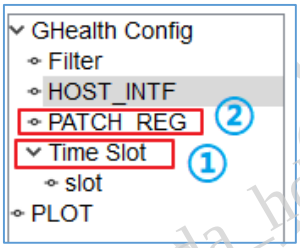


Figure 5-6 Outline annotation

Table 5-2 Outline

| UI ID | Type | Description |
|---|---|---|
| ❶ | Root topic | Config topic with subtopic which supports folding up or unfold |
| ❷ | Leaf topic | Config topic with no subtopic |

Table 5-3 Outline operations

| Function | Operation | Description |
|---|---|---|
| Switch Page | Single click item | Replace config page to select topic |
| Expand or collapse subtopic | Click popup icon or double click item | Expand or collapse subtopic (if exist) belonging to this topic |
| Content Check | Switch between pages or modify some key config items | - |

## 5.3 Config Operations

Figure 5-7 Config operations annotation

Table 5-4 Config operation manual

| UI ID | Type | Description | Note |
|---|---|---|---|
| ❶ | Pushbutton | Save Config file by windows file manager | - |
| ❷ | Pushbutton | Load config file from local by windows file manager | - |
| ❸ | Pushbutton | Start to test the config present in UI immediately; this function is useful when testing the effect of some config items.<br>Disappear when no device is connected. | This function can automatically process the following steps:<br>1. Temporarily save current file<br>2. stop sampling (if sampling is running) |

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| | | | |
|---|---|---|---|
| | | | 3. resend config file<br>4. restart sampling |

## 5.4 Controls

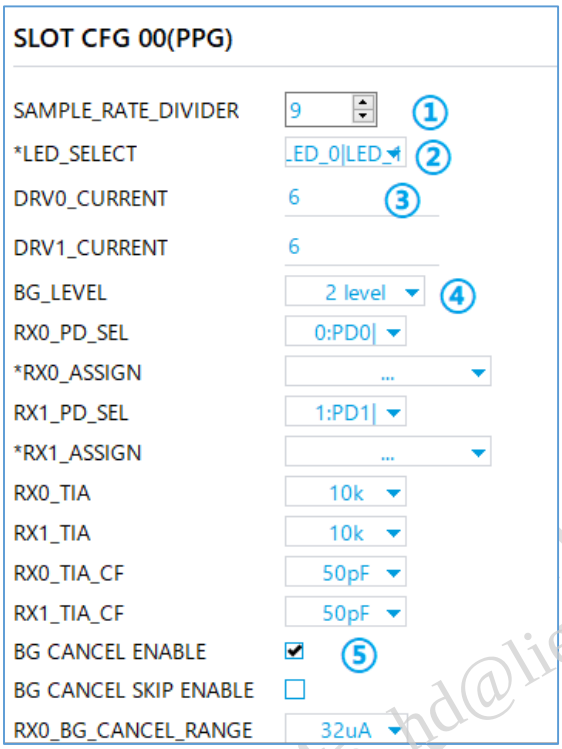


Figure 5-8 Controls

Table 5-5 Controls manual

| UI ID | Type | Description | Note |
|---|---|---|---|
| 1 | Spinbox | Integer number input | If the name content contains HEX, it indicates that hex-adacimal number should be filled |
| 2 | Multicombobox | Multiple choices | Tooltip shows all items selected (shows up when cursor covers) |
| 3 | LineEdit | Float number input | - |
| 4 | Combobox | Single choice | - |
| 5 | Checkbox | Boolean choice | - |

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## 5.5 PLOT

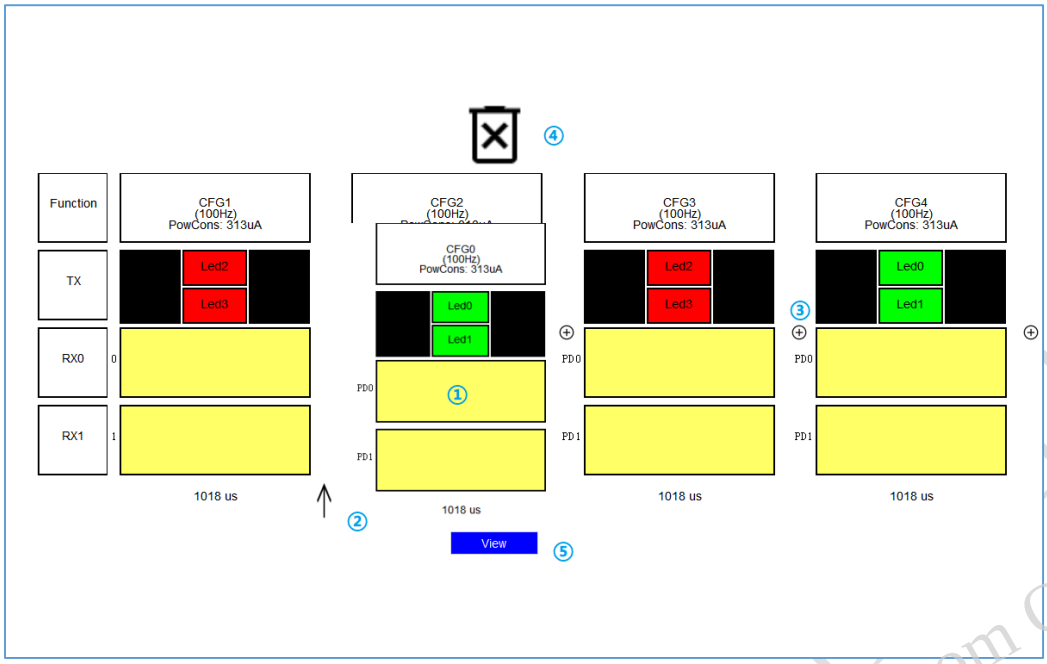


Figure 5-9 Plot special page annotation

Table 5-6 Plot special page manual

| UI ID | Type | Description | Note |
|---|---|---|---|
| ❶ | Label | Select slot, left cursor long press on a slot | Only PPG slot supports right click menu and single click to use PPG slot wizard to set slot config; please refer to section 5.5.1 PPG slot wizard for details.<br>The PPG slot wizard page will disappear when it loses mouse focus (such as when you click other window or other window shows up). |
| ❷ | Label | Indicate position where the slot will be inserted | Always insert to the indicated slot |
| ❸ | Pushbutton | Add slot here | - |
| ❹ | Label | Indicate the delete slot operation. If the slot is on the remove position, this label becomes red | Move slot out of page means to delete this slot |
| ❺ | Pushbutton | Show up a view window supporting zoom | Do not support any operation in view |

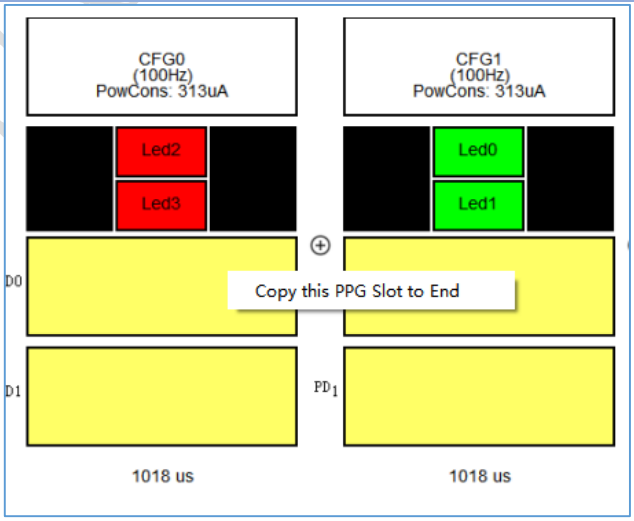


Figure 5-5 Plot PPG slot right menu

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### 5.5.1 PPG slot wizard

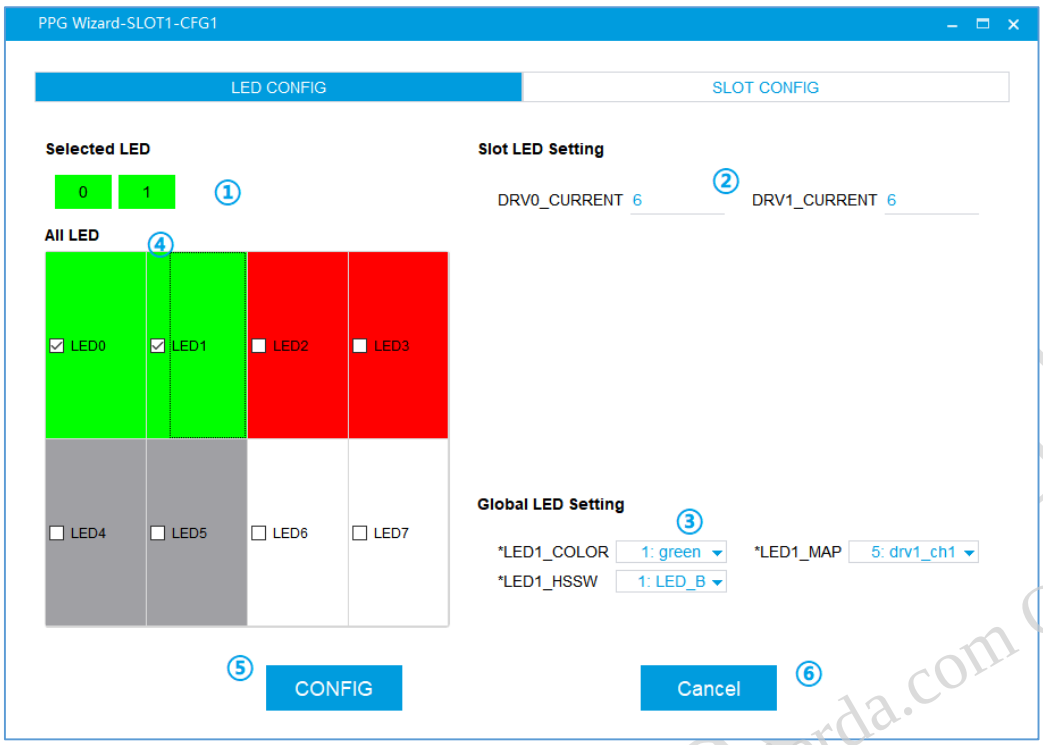


Figure 5-6 PPG slot wizard-LED CONFIG

Figure 5-7 PPG slot wizard- SLOT CONFIG

Table 5-7 PPG slot wizard annotation

| UI ID | Type | Description | Note |
|---|---|---|---|
| ❶ | Label list | Show all LEDs selected in this slot | - |
| ❷ | Config item list | Show the information about the slot | - |
| ❸ | Config item list | Show the information about the selected LED in ID❹ | Any modification may influence other slot |
| ❹ | Checkbox table | All LED channels that chip supports are arrayed as a table, user can select the required LED by ticking the box | Any check or click will change the target of Global LED setting |

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| | | | |
|---|---|---|---|
| ❺ | Pushbutton | Commit all modifications and quit PPG slot wizard | - |
| ❻ | Pushbutton | Give up all modifications and quit PPG slot wizard | - |
| ❼ | Label | Represent LED work flow of this slot.<br>Follow selected LED in ID1 and BG_LEVEL in ID9 to change pixmap | - |
| ❽ | Label | Click to switch the selected PD | Picture will vary according to RXx_PD_SEL in ID10 |
| ❾ | Config item list | Show the information about the PD selected in ID8 | Empty at the beginning until first PD is clicked |
| ❿ | Config item list | Show the information about the slot | Contain all config of PPG slot |

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# 6 DataProcess

## 6.1 Overview

The DataProcess module shows data processing operations (such as filters or GHealth algorithms) applied in the configuration and user can enable and disable the existing operation manually.

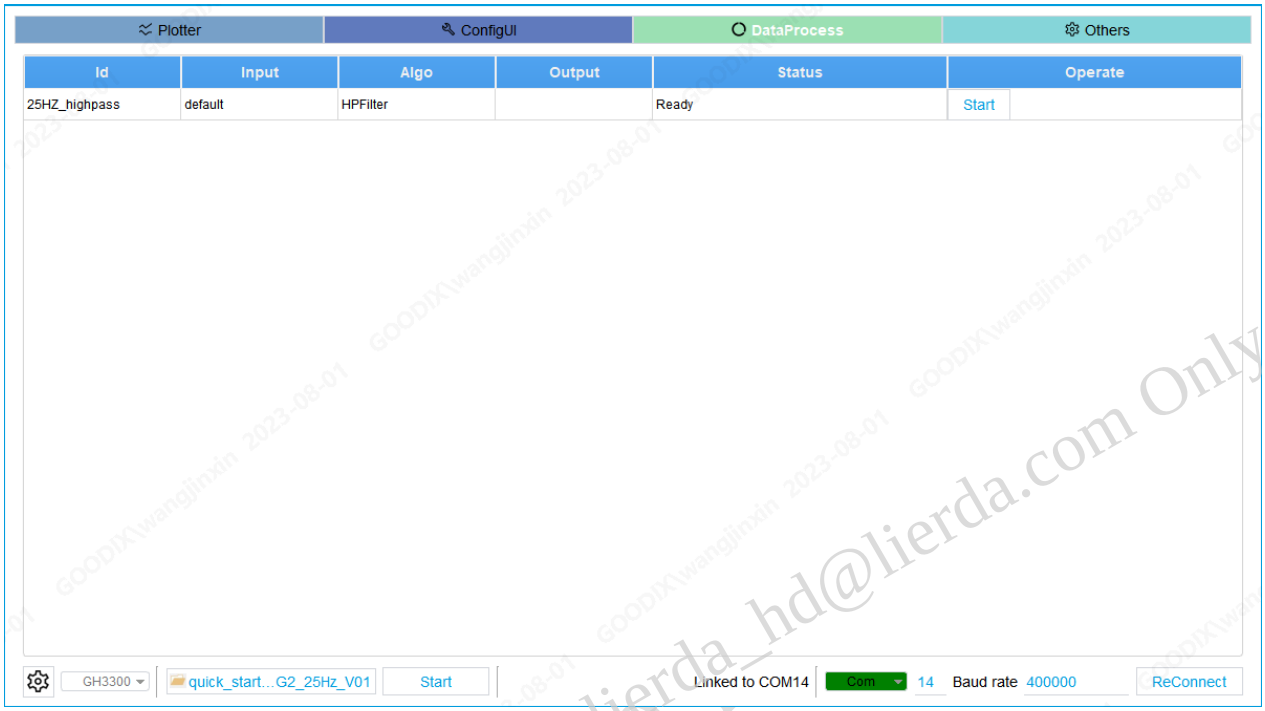


Figure 6-1 DataProcess overview

- Users can also enable or disable the data processing by right clicking the file frame in Plotter interface.

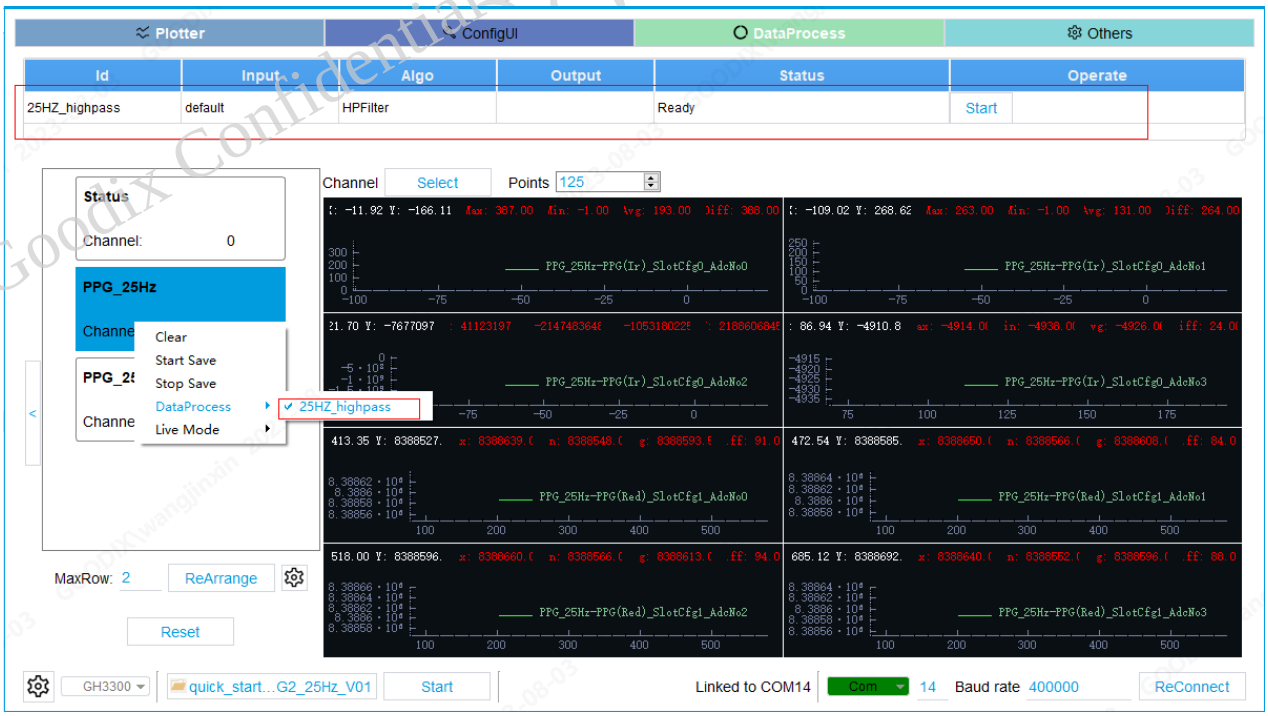


Figure 6-2 Enable data processing through file frames

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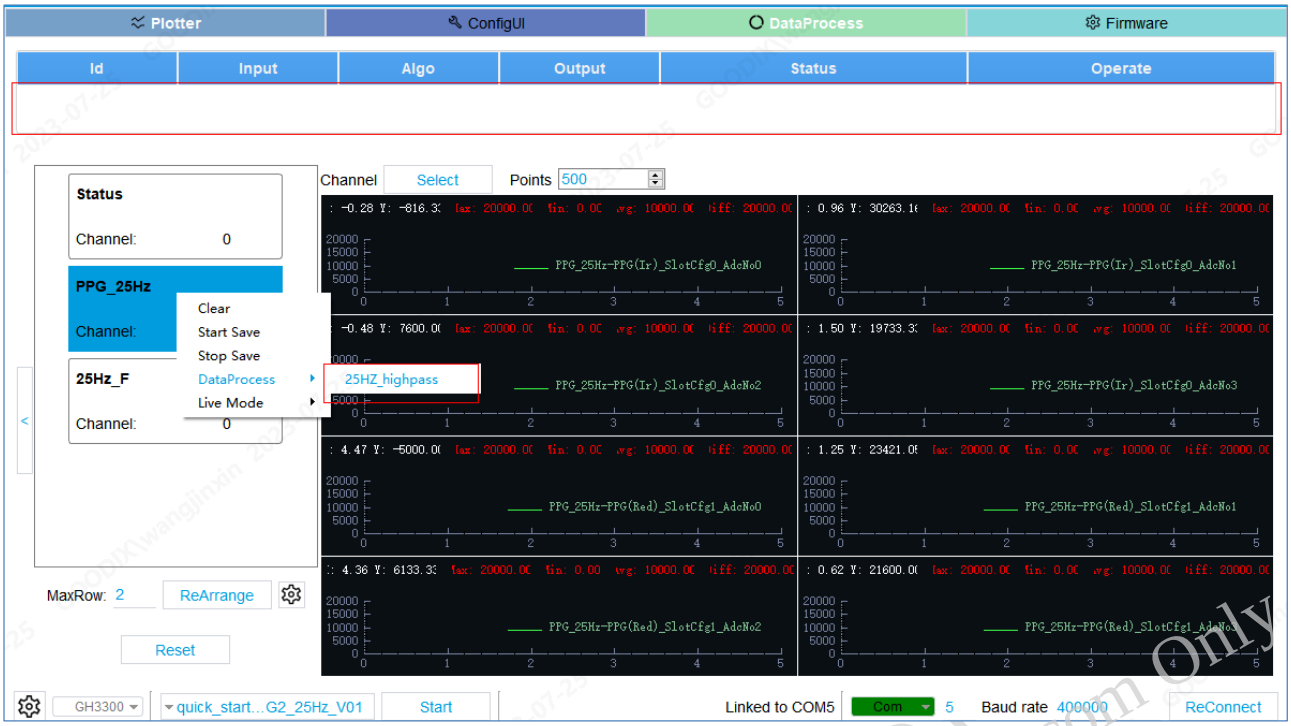


Figure 6-3 Disable data processing through file frames

## 6.2 Main page

| Id | Input | Algo | Output | Status | Operate |
|---|---|---|---|---|---|
| 100HZ_highpass | default | HPFilter | | Ready | Start |

Figure 6-4 DataProcess entry table

Table 6-1 DataProcess manual

| Table | Type | Description |
|---|---|---|
| ID | label | Present name of this process |
| Input | label | Data source |
| Algo | label | Algorithm name |
| Status | Label | Running status |
| Operate | Pushbutton | Switching running state of algorithm |

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# 7 Others

## 7.1 Overview

The Others module provides interface to communicate with firmware directly.

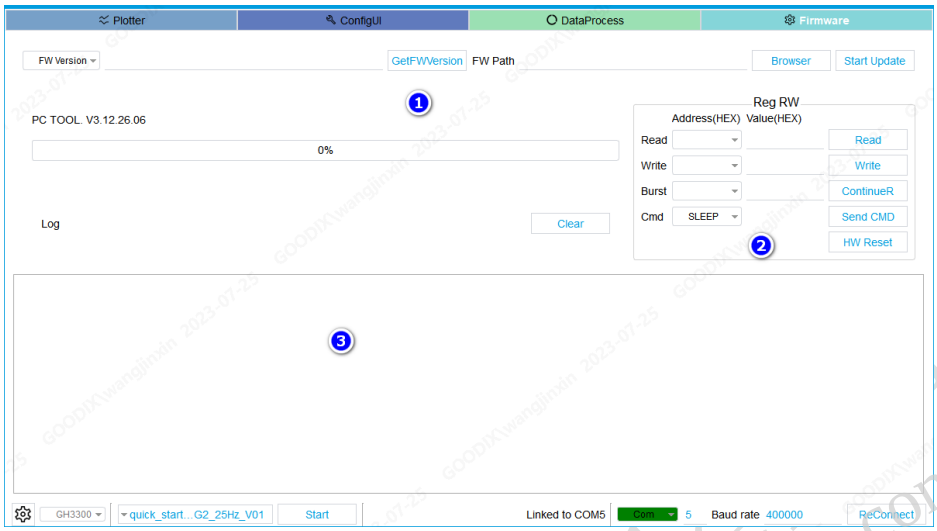


Figure 7-1 Others

1. Check version of firmware or update firmware using the selected bin file.

Figure 7-2 version& update(in update)

2. Set specified cmd or register address to interact with Goodix health sensor directly.

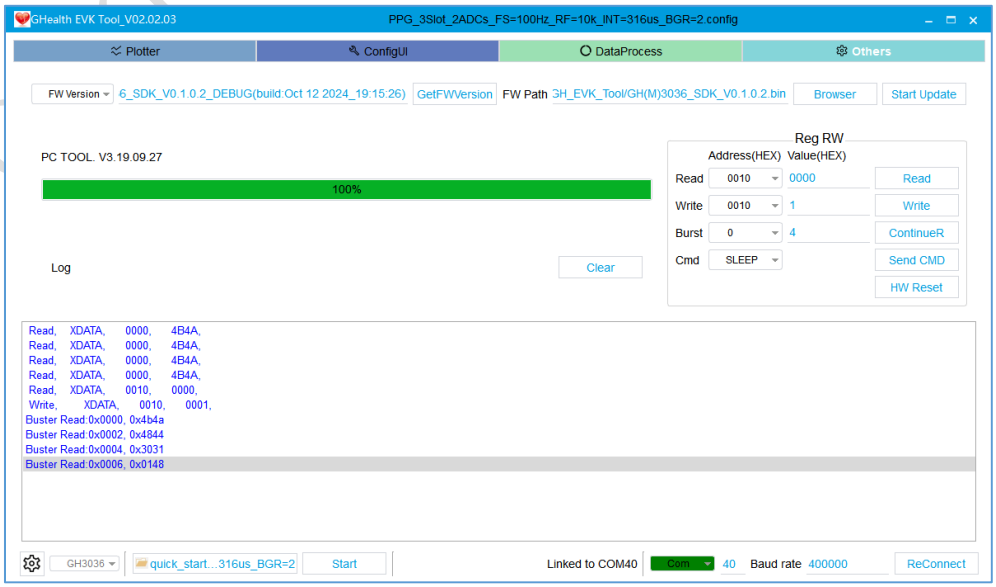


Figure 7-3 Chip control usage

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3. Print return information in log text editor.

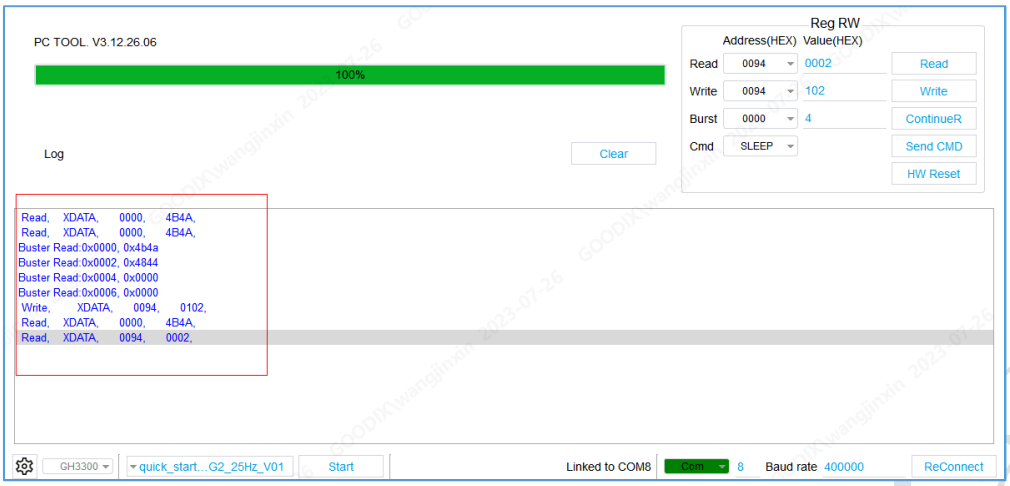


Figure 7-4 log information text editor

## 7.2 Version& Firmware Update

Figure 7-5 Version& Firmware Update

Table 7-1 Version& Firmware Update manual

| UI ID | Type | Description | Note |
|---|---|---|---|
| 1 | Combobox | Select the version type that you want to check | - |
| 2 | lineedit | Show the version information | Allow to edit to check all info if version string is too long |
| 3 | Pushbutton | Get the version according to the version type | - |
| 4 | Lineedit | Show the selected version information | - |
| 5 | Pushbutton | Show a view window supporting zoom to select the file that needs to be upgraded | - |
| 6 | Pushbutton | Start to update firmware with the selected bin file. | Block all UIs in this module during the update. Do not perform any device operations during the update process, as this may lead to an update failure. |
| 7 | Processbar | Indicate the update progress | Print log continually in the update process |

### 7.2.1 Auto Firmware Update

The firmware version will be checked when the EVK board is connected to the EVK PC Tool for the first time and the old firmware will be automatically updated.

1. Auto firmware update only supports serial connection, and does not support Bluetooth connection.
2. Insert serial port and wait for about 10 seconds to recognize the old firmware.

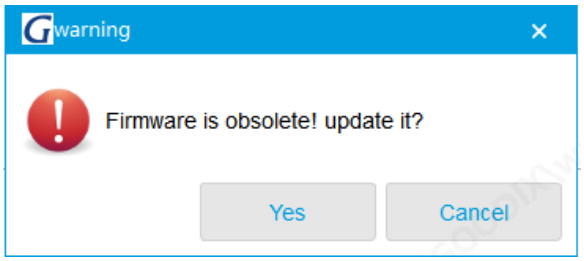


Figure 7-6 Automatically version check when device is inserted

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3. Click **Yes** on the pop up page to start updating, wait until the update is complete and the blue LED in EVK board remains on.
4. Once the update is successful, reconnect to the EVK board to make the new firmware work.
5. Click **Cancel** to keep using the old firmware, but the tool only supports getting firmware version or updating in this case.

## 7.3 Chip Control

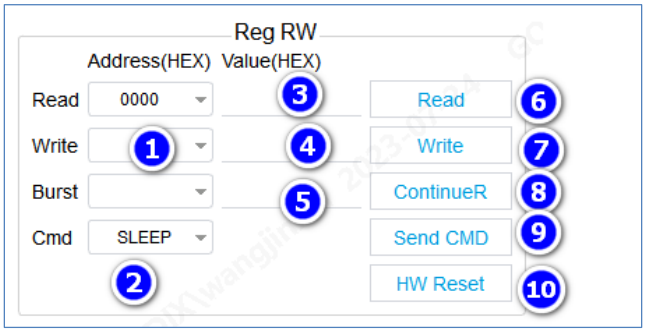


Figure 7-7 Chip Control annotation

Table 7-2 Chip Control manual

| UI ID | Type | Description | Note |
|---|---|---|---|
| ❶ | Inputbox | Input hex register address | - |
| ❷ | Combobox | Choose operations to be performed on the chip | - |
| ❸ | Lineedit | Show the register value | Hex |
| ❹ | Lineedit | Input the register value that needs to be written to the chip | Hex |
| ❺ | Pushbutton | Input the number of registers that need to be read at one time | No more than 20.<br>Read from the written register address |
| ❻ | Pushbutton | Read single register value from the chip | Show in log dialog |
| ❼ | Pushbutton | Write register value to the chip | Show in log dialog |
| ❽ | Pushbutton | Read multiple register values from the chip | Show in log dialog |
| ❾ | Pushbutton | Send cmd to the chip | - |
| ❿ | Pushbutton | Restart chip by powering off and then back on | - |

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# 8 Appendix

## 8.1 Specification

Table 8-1 Specification of EVK PC Tool

| Item | Specification |
|---|---|
| Max communication speed in serial port | 400K baud rate, 25kByte/s in actual fact |
| Max file Writing speed | 5Mbyte/s |
| Max plot data | 3(plot)*20K or 12(plot)*1K |

## 8.2 System Requirements

Table 8-2 System Requirements

| Item | Description | Note |
|---|---|---|
| PC OS | Windows 7 or Windows 10 | Do not support Win11 or XP |
| PC RAM | ≥4GB | - |
| Screen | 768P to 2160P | 100% - 150% Zoom |
| Disk Space | ≥1GB | Space that program and data file will occupy. |
| Runtime library | MS VC++ 2013&2015&2017 redistributable x86<br>directX 9 and later | Recommend to use **DirectX Repair** to repair runtime library problem. |
| Driver | CH343 serial driver (old version EVK board)<br>CP21xx serial driver (old version uses BLE dongle)<br>J-link CDC serial driver (current version) | Usually only CDC driver is needed;<br>Windows7 may lack CDC driver; please install Jlink by yourself or contact Goodix staff for assistance;<br>Jlink: https://www.segger.com/downloads/jlink |

## 8.3 Known Issues

Table 8-3 Known issues

| ID | Description | Solution |
|---|---|---|
| 1 | Cannot input to application | System input editor compatibility problem; use origin English input editor to do some input |
| 2 | Fail to install runtime library, and fail to launch the program | Go to folder named **driver** under application install folder, install **vc_redist_x86.exe** and **vcredist_x86_2013.exe** or install according to Table 8-2 System Requirements |
| 3 | Fail to install driver, and cannot connect to device | Solve the problem according to Table 8-2 System Requirements item **Driver** by manually installing drivers |
| 4 | EVK board does not work after disconnection during sampling | Try to power off EVK board and reconnect to EVK PC Tool |
| 5 | Cannot read firmware version after sampling | Cutoff power supply for 5s, and then reboot EVK board |

## 8.4 File Format

Table 8-4 Common headers

| Header | Description |
|---|---|
| Timestamp | Unit : millisecond |
| FRAME_ID | Start from 0 and increment by 1 each time, stopping only if a packet is lost during transmission. |

### 8.4.1 PPG format

Table 8-5 PPG Data valid header

| Header | Description |
|---|---|
| ACC | G-sensor data |
| FLAG0 | Auto adjustment status<br>bit0: If Channel0 is in adjustment. 0:no; 1: yes.<br>bit1: If Channel1 is in adjustment. 0:no; 1: yes. |

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| Header | Description |
|---|---|
| | …<br>Bit15: If Channel15 is in adjustment. 0:no; 1: yes. |
| FLAG1 | Saturation status<br>bit0: If Channel0 mixdata is oversaturated. 0:no; 1: yes.<br>bit1: If Channel1 mixdata is oversaturated. 0:no; 1: yes.<br>…<br>Bit15: If Channel15 mixdata is oversaturated. 0:no; 1:yes. |
| FLAG2 | bit0: Division by zero exception. 0:no 1:yes |
| FLAG3 | Parameter update flag.<br>bit0: Channel0 parameter has been updated. 0:no 1: yes<br>bit1: Channel1 parameter has been updated. 0:no 1: yes<br>…<br>bit15: Channel15 parameter has been updated. 0:no 1: yes |
| FLAG4 | BG cancel share flag.<br>bit0: Channel0 BG cancel parameter has been shared. 0:no 1: yes<br>bit1: Channel1 BG cancel parameter has been shared. 0:no 1: yes<br>…<br>bit15: Channel15 BG cancel parameter has been shared. 0:no 1: yes |
| FLAG5 | DRE update flag.<br>bit0: Channel0 DRE update. 0:no 1: yes<br>bit1: Channel1 DRE update. 0:no 1: yes<br>…<br>bit15: Channel15 DRE update. 0:no 1: yes |
| Slotcfg(x)_<br>rx(x) | PPG channel x (24bit) |
| slotcfg(x)_<br>rx(x)_agcinfo | [3:0] Gain<br>[5:4] Bg cancel-range<br>[7:6] DC cancel-range<br>[15:8] DC cancel code<br>[29:16] sum of LED drivers current (unit : 0.1mA) |
| slotcfg(x)_<br>rx(x)_ledinfo | [11:0] led driver0 current (unit: 0.1mA)<br>[23:12] led driver1 current (unit: 0.1mA) |
| slotcfg(x)_<br>rx(x)_ physics | PPG channel x<br>PD current (unit: pA, 32 bits) |

### 8.4.2 PPG BG format

Table 8-6 PPG BG Data valid header

| Header | Description |
|---|---|
| Slotcfg(x)_<br>rx(x)_bg_data | PPG channel x background sample data (24-bit) |

### 8.4.3 PPG MIX data format

Table 8-7 PPG BG Data valid header

| Header | Description |
|---|---|
| Slotcfg(x)_<br>rx(x)_ mixdatax | PPG channel x LED sample data (24-bit) |

### 8.4.4 PPG DRE data format

Table 8-8 PPG BG Data valid header

| Header | Description |
|---|---|
| Slotcfg(x)_<br>rx(x)_ dredatax | PPG channel x DRE data (24-bit) |

### 8.4.5 PPG parameter format

Table 8-9 PPG BG Data valid header

| Header | Description |
|---|---|
| Slotcfg(x)_<br>rx(x)_ paradata | PPG channel x parameters (24-bit)<br>[1:0] DC cancel range |

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| Header | Description |
|---|---|
| | [3:2] BG cancel range<br>[6:4] TIA gain<br>[7:7] BG Cancel skip flag<br>[15:8] DC cancel code<br>[23:16] BG cancel code |

### 8.4.6 CAP format

Table 8-10 CAP Data valid header

| Header | Description |
|---|---|
| Slotcfg(x) | Capacitance data |

Please note that GH3038 does not support capacitance measurement.

### 8.4.7 DRE-Dc-info data format

Table 8-7 PPG Data valid header

| Header | Description |
|---|---|
| ACC | G-sensor data |
| FLAG0 | Auto adjustment status<br>bit0: If Channel0 is in adjustment. 0:no; 1: yes.<br>bit1: If Channel1 is in adjustment. 0:no; 1: yes.<br>…<br>Bit15: If Channel15 is in adjustment. 0:no; 1: yes. |
| FLAG1 | Saturation status<br>bit0: If Channel0 mixdata is oversaturated. 0:no; 1: yes.<br>bit1: If Channel1 mixdata is oversaturated. 0:no; 1: yes.<br>…<br>Bit15: If Channel15 mixdata is oversaturated. 0:no; 1:yes. |
| FLAG2 | bit0: Division by zero exception. 0:no 1:yes |
| FLAG3 | Parameter update flag.<br>bit0: Channel0 parameter has been updated. 0:no 1: yes<br>bit1: Channel1 parameter has been updated. 0:no 1: yes<br>…<br>bit15: Channel15 parameter has been updated. 0:no 1: yes |
| FLAG4 | BG cancel share flag.<br>bit0: Channel0 BG cancel parameter has been shared. 0:no 1: yes<br>bit1: Channel1 BG cancel parameter has been shared. 0:no 1: yes<br>…<br>bit15: Channel15 BG cancel parameter has been shared. 0:no 1: yes |
| FLAG5 | DRE update flag.<br>bit0: Channel0 DRE update. 0:no 1: yes<br>bit1: Channel1 DRE update. 0:no 1: yes<br>…<br>bit15: Channel15 DRE update. 0:no 1: yes |
| Slotcfg(x)_<br>rx(x) | DRE-Dc-info data channel x (24bit) |
| slotcfg(x)_<br>rx(x)_agcinfo | [3:0] Gain<br>[5:4] Bg cancel-range<br>[7:6] DC cancel-range<br>[15:8] DC cancel code<br>[29:16] sum of LED drivers current (unit : 0.1mA) |
| slotcfg(x)_<br>rx(x)_ledinfo | [11:0] led driver0 current (unit: 0.1mA)<br>[23:12] led driver1 current (unit: 0.1mA) |
| slotcfg(x)_<br>rx(x)_ physics | Reserve , value is 0 |

## 8.5 Transform formulation

Table 8-8 Transform formulations

| Function | Formulation | Note |
|---|---|---|

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| | | |
|---|---|---|
| Rawdata to pd current | Ipd=(rawdata)/2^24* Vref /gain*1000 | Unit: mA<br>Vref: Typical 1.8V (modifiable)<br>Gain:<br>refer to Table 8-9 PPG gain range |

Table 8-9 PPG gain range

| Order | Resistance(KΩ) |
|---|---|
| 0 | 10 |
| 1 | 25 |
| 2 | 50 |
| 3 | 100 |
| 4 | 250 |
| 5 | 500 |
| 6 | 1000 |

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